

Department Overview

- **Department Name:** Department of Software Engineering
- **Faculty:** Faculty of Information & Communication Technology (FICT)
- **Accreditation:** The program is accredited by the **Pakistan Engineering Council (PEC)**.
- **Core Focus:** Combines theoretical and practical software engineering principles. It encompasses the systematic, disciplined, and quantifiable approach to the design, development, operation, and maintenance of large-scale, reliable, efficient, and economical software systems.
- **History:** The department has been offering a four-year undergraduate software engineering degree since 2012 to meet growing national and international market demands.
- **Specialization Fields:** Requirements engineering, quality engineering, software testing, and project management.

Vision and Mission

- **Vision:** To attain excellence in education and research at the national and international level and produce technically, professionally, and ethically sound graduates.
- **Mission:** To produce the next-generation Software Engineering graduates who are fully equipped with concrete software engineering knowledge, real-world problem-solving skills, research abilities, excellent leadership qualities, and entrepreneurial abilities required for their future careers, and serving their communities.

Bachelor of Science in Software Engineering (BSSE)

Admission Requirements

- **Academic Background:** F.Sc (Pre-Engineering) / ICS with Mathematics and Physics from any recognized board (or equivalent qualification) with at least **60% marks**. Alternatively, a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of **60% marks**.

Degree Requirements

- **Total Credit Hours:** 132
- **Course Structure:** 42 courses plus the Final Year Design Project (FYDP Part I & II).
- **Minimum Academic Standing:** CGPA $\geq$ 2.0

Program Educational Objectives (PEOs)

The department targets training students across all aspects of the software development lifecycle to achieve the following career and professional objectives:

1. Utilize critical-thinking and engineering abilities to excel in the domain of Software Engineering.
2. Act as professional, ethical, and knowledgeable citizens.
3. Demonstrate leadership and work collaboratively in diverse teams and organizations.

Curriculum Plan (Program Schema by Semester)

Note: Credit hour listings follow the format (Theory-Lab).

Semester	Course Code	Course Title	Prerequisite	Credit Hours
1st	IT-102	Information and Communication Technology (ICT)	NIL	2-1
	SE-2XX	Software Engineering Fundamentals	NIL	3-0
	ENG-161	Reading and Writing Skills	—	3-0
	PHY-205	Applied Physics	NIL	2-1
	HUM-138 / HUM-1XX	Psychology / Islamic Studies and Ethics	NIL / NIL	2-0 / 2-0
	XX-1XX	Occupational Health and Safety	NIL	1-0
2nd	CS-114	Programming Fundamentals	NIL	3-1
	ENG-361	Communication & Presentation	NIL	3-0

		Skills		
	MATHA-234	Discrete Structures	NIL	3-0
	MATHP-118	Calculus & Analytical Geometry	NIL	3-0
	CS-119 / HUM-102	Web Development (WD) / Pakistan Studies & Global Perspective	NIL / NIL	2-1 / 2-0
3rd	CS-212	Object-Oriented Programming	Programming Fundamentals	3-1
	STAT-101	Probability & Statistics	NIL	3-0
	CE-2XX	Computer Architecture and Logic Design	NIL	3-1
	MATHP-111	Linear Algebra	NIL	3-0
	CE-302	SE Elective I (Web Engineering (WE))	Web Development	2-1

4th	CS-214	Data Structures and Algorithms (labeled as OOP in page text)	Object-Oriented Programming	3-1
	SE-231	Software Design & Architecture	Software Engineering Fundamentals	2-1
	IT-3XX	Human-Computer Interaction	NIL	2-1
	XX-XXX	SE Elective – II	NIL	3-0
	MGMT-206 / MATHA-213	Principles of Management / Complex Variables & Transform	NIL / NIL	3-0 / 2-0
5th	SE-332	Software Construction and Development	Software Design & Architecture	2-1
	MGMT-2XX	Entrepreneurship and Marketing	NIL	2-0
	XX-XXX	SE Elective – III	NIL	3-0

	CS-332 / CS-341	Database Systems / Operating Systems	NIL / NIL	3-1 / 3-1
6th	SE-321	Software Quality Engineering	Software Engineering Fundamentals	3-0
	TE-307	Computer Networks	NIL	3-1
	MATHA-216	Numerical Analysis	NIL	2-1
	XX-XXX	MDEE- I	NIL	3-0
	HUM-266	Technical & Business Writing	NIL	3-0
7th	SE-352	Software Project Management	NIL	3-0
	IT-321	Information Security	NIL	3-0
	SE-491L	Senior Design Project – I (FYDP-I)	NIL	0-3
	XX-XXX	SE Elective – IV	NIL	3-0

	CS-4XX	Design and Analysis of Algorithms	Data Structures and Algorithms	3-0
8th	XX-XXX	MDEE-II	NIL	3-0
	SE-492L	Senior Design Project – II (FYDP-II)	Senior Design Project – I	0-3
	XX-XXX / XX-XXX	SE Elective – V / SE Elective – VI	NIL / NIL	3-0 / 3-0
	SE-471	Formal Methods in Software Engineering	NIL	3-0